LAB : 13

Title : Structure in C++

► Objectives

- To understand the concept of structures in C++.
- To learn how to declare and initialize structures.
- To practice accessing structure members using dot notation.
- To apply structures for storing and displaying real-world data.

► Theory

- A structure is a collection of variables (members) of different data types grouped under one name.
- **Syntax:**

```
struct StructName {
    int id;
    float value;
    char name[20];
};
```

- Unlike arrays (same type only), structures can hold mixed types.
- Structures can be nested, copied, and passed to functions.

- Members are accessed using the dot operator **(.)**.

► Lab Tasks

Task 1: Structure Declaration and Initialization

```
#include <iostream>      // tell the compiler to include input output stream library
using namespace std;     // tells the compiler to use standard namespace

struct Person {          // structure declaration
    string name;         // member: name
    int age;             // member: age
    string address;      // member: address
};

int main() {                                             // main function starting point
    Person person1 = {"Naeem ul Haq", 19, "Peshawar"};   // initialize structure
    cout << "Name: " << person1.name << endl;            // display name
    cout << "Age: " << person1.age << endl;              // display age
    cout << "Address: " << person1.address << endl;      // display address
```

```
    return 0;                    // tells the compiler that progrsm has ended successfully
}
```

Task 2: Student Structure

```
#include <iostream>       // tell the compiler to include input output stream library
using namespace std;      // tells the compiler to use standard namespace

struct Student {           // structure declaration
    string name;           // member: name
    int age;               // member: age
    char grade;            // member: grade
};
int main() {                                          // main function starting point
    Student student1 = {"Ali Khan", 20, 'A'};                 // initialize student
    cout << "Name: " << student1.name << endl;                // output message
    cout << "Age: " << student1.age << endl;                  // output message
    cout << "Grade: " << student1.grade << endl;              // output message
    return 0;          // tells the compiler that program has ended successfully
}
```

Task 3: Accessing Structure Elements

```
#include <iostream>         // tell the compiler to include input output stream library

using namespace std;             // tells the compiler to use standard namespace


struct Book {                               // structure declaration

   string title;                            // member: title

   string author;                           // member: author

   float price;                             // member: price

   int pages;                               // member: pages

};


int main() {                               // main dunction starting point

 Book book1 ={"C++ Programming", "Bjarne Stroustrup", 999.99, 1200};    //struc1

 Book book2 = {"Data Structures", "Mark Weiss", 750.50, 850};              //struc2


   cout << "Book 1: " << book1.title << ", " << book1.author           //output message

       << ", Price: " << book1.price << ", Pages: " << book1.pages << endl;


   cout << "Book 2: " << book2.title << ", " << book2.author           //output message

       << ", Price: " << book2.price << ", Pages: " << book2.pages << endl;
```

```
    return 0;              // tells the compiler that program has ended successfully
}
```

► Results

- Successfully declared and initialized structures with multiple members.
- Demonstrated accessing structure members using dot notation.
- Showed how structures can store different types of data together.
- Applied structures to real-world examples (Person, Student, Book).